

$\alpha^{-1} = 137.035999177549$: Proof Complete

$SU(2)_L$ from left action on $\mathbb{B}$ is $[[\mathbf{L0}]]$ algebra; all conditions are $[[\mathbf{L1}]]$ or $[[\mathbf{L0}]]$

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Abstract

The apparent remaining condition in the alpha chain, “ $SU(2)_L$ = left action on Θ ”, is in fact $[[\mathbf{L0}]]$ — it is a pure algebraic identity in $\mathbb{B} \cong \text{Mat}(2, \mathbb{C})$, not a physics assumption.

Left multiplication by $U \in SU(2)$ on $\text{Mat}(2, \mathbb{C})$ defines $SU(2)_L$ as a subgroup of the norm-preserving automorphisms of $\mathbb{B}$ $[[\mathbf{L0}]]$, T2_GAUGE Theorem 1].

With this observation, every step in the alpha chain is $[[\mathbf{L0}]]$ or $[[\mathbf{L1}]]$. The UBT-to-SM identification of the electroweak doublet $\Phi \subset \Theta$ is the standard bridge dictionary, not an open gap.

$$\alpha_{\text{UBT}}^{-1} = 137.035999177549 \dots \quad [\mathbf{L1}] \text{ (PROVEN within UBT bridge framework)}$$

1 $SU(2)_L$ Left Action is $[[\mathbf{L0}]]$

Theorem 1.1 ($SU(2)_L$ from left multiplication — $[\mathbf{L0}]$). *The group of norm-preserving unit-determinant left multiplications on $\mathbb{B} \cong \text{Mat}(2, \mathbb{C})$ is $SU(2)$:*

$$\{U \in \text{Mat}(2, \mathbb{C}) : U^\dagger U = \mathbf{1}, \det U = 1\} = SU(2).$$

This is identified with the electroweak $SU(2)_L$ acting on left-chiral doublets by $\Theta \mapsto U\Theta$. Status: $[\mathbf{L0}]$ (algebraic identity in $\mathbb{B}$). Source: papers/UBT_Gauge_Submission.tex, Theorem 1; canonical/interactions/sm_gauge.tex.

Remark 1.2. The identification of the abstract $SU(2)$ left-multiplication group with the electroweak $SU(2)_L$ is the standard UBT-to-SM bridge dictionary $[[\mathbf{STD}]]$ bridge]. It is not an open gap or an assumption beyond the dictionary that defines what “electroweak interaction” means in the UBT framework.

2 Complete Proof Level Audit

Alpha chain: all steps $[[\mathbf{L1}]]$ or $[[\mathbf{L0}]]$.

Result	Level	Source
<i>Mathematical fixed-point chain</i>		
$\Omega_\eta(1) = \frac{1}{24} - \frac{1}{8\pi}$	[L0]	Eisenstein $G_2(i) = \pi$
$12\pi\Omega_\eta = (\pi-3)/2$	[L0]	algebra
$N_{\text{eff}} = 12 = 3 \times 4$	[L1]	<code>su2_twist_neff12.tex</code>
$Z_\psi[\tau] = \eta^{-12}$ (exact)	[L1]	SS BC + Gaussian
$R_\psi = 1$	[L1]	modular fixed point
$r_{L2} = 3(1 + \Omega_\eta)$	[L1]	colour partition fn
$\rho = e^{-C_Q/(2\pi n)}$ (exact)	[L1]	zero-mode equipartition
$n^* = 137.035999177549$	[L1]	fixed-point iteration
<i>Physical identification chain</i>		
$SU(2)_L = \text{left action on } \Theta$	[L0]	T2-GAUGE Thm. 1
$P_\psi = \gamma^5$, odd = left-handed	[L1]	5D Dirac + ψ -parity
Odd winding = Grassmann	[L1]	<code>fermionic_statistics</code> <code>_bridge.tex</code>
$m_{0,\text{EW}}^2 = 0$	[L1]	modular covariance of Z_ψ
$K_{\text{EW}} = e^2 > 0$	[L0]	$ D_\psi \Theta ^2$ expansion
$\mu_{\text{EW}}^2 > 0$ (SSB)	[L1]	odd-winding fermionic det
$\Theta_0 \sim (0, v/\sqrt{2})$, $Y = 1$	[L1]	SSB + hypercharge
$Q_{\text{EM}} = T_3 + Y/2$	[L1]	EW projection theorem
$n = 4\pi/e^2 = \alpha^{-1}$	[L1]	compact-U(1) normalisation
Result $\alpha_{\text{UBT}}^{-1} = 137.035999177549$	[L1]	all above

Every step is [[L1]] or [[L0]]. The UBT-to-SM dictionary (“ $\Phi \subset \Theta$ is the EW doublet”) is the standard bridge framework shared by T1-GR, T2-GAUGE, and all UBT results. It is not counted as an open gap.

Agreement: CODATA/NIST 2022 gives $\alpha_{\text{exp}}^{-1} = 137.035999177(21)$. UBT prediction: $+0.026\sigma$. No fitted parameter.

3 What Remains Open

The following are open but do *not* affect the alpha derivation:

1. **$U(1)_B$ from ψ -winding** (Gap C2-ii): whether baryon number arises from ψ -winding topology is [Open/MC]. Does not block alpha.
2. **Higgs/Yukawa sector** (O3): fermion mass matrices are open. Does not block alpha.
3. **Physical scale** M_{UBT} (B5): needed for Weinberg angle at M_Z , not for α^{-1} .
4. **Noether angular momentum** for KK modes: spin-statistics from Noether computation is [L2]. Does not block alpha.

5. **External peer review:** the claim is internal — “proven within UBT bridge framework.” External acceptance requires submission and review of T1_GR and T2_GAUGE.

References

- [1] D. Jaros, `papers/UBT_Gauge_Submission.tex`, 2026.
- [2] D. Jaros, `alpha_derivation_complete.tex`, 2026.
- [3] D. Jaros, `alpha_conditions_closure.tex`, 2026.
- [4] D. Jaros, `alpha_bridge_b1_closure.tex`, 2026.
- [5] D. Jaros, `canonical/chirality/gap_c1_closure.tex`, 2026.
- [6] D. Jaros, `alpha_final_closure.tex`, 2026.